A D PATEL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

THE CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY

CANTEEN FOOD ORDERING SYSTEM

PYTHON MINI PROJECT

A desktop application for menu selection, cart management and billing

FOOD MENU | SHOPPING CART | MYSQL DATABASE | BILL RECEIPT

PROJECT SUBMITTED BY

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PROJECT OVERVIEW & OBJECTIVES

PROJECT OVERVIEW

A Python desktop application that lets canteen users browse food items, choose quantities, manage a cart and generate a bill through a graphical interface.

PROBLEM ADDRESSED

Manual order taking and billing can be slow and error-prone. This system organizes menu details, calculates totals and stores confirmed orders.

PROJECT OBJECTIVES

- Make ordering simple and user-friendly.
- Display food items, categories and prices.
- Calculate the total bill automatically.
- Store order and item details in MySQL.

EXPECTED OUTCOME

- Clear menu and shopping-cart view.
- Capture customer name and payment method.
- Generate a receipt after order confirmation.
- Save a text-file copy of the bill.

TECHNOLOGY STACK

PYTHON

Core language for application logic, calculations and database operations.

TKINTER

Builds the desktop GUI with windows, buttons, labels, input fields and tables.

MYSQL

Stores menu items, order records and the items included in each order.

MYSQL CONNECTOR

mysql-connector-python connects the Python application to MySQL.

OOP & DICTIONARIES

Classes separate responsibilities; dictionaries hold menu and cart data.

EXCEPTION HANDLING

Catches database/file errors and validates quantities with helpful messages.

Required package: mysql-connector-python

MAIN FEATURES

View Menu

Food name, category and price load from MySQL.

Add to Cart

Select a food item and enter a positive whole-number quantity.

Manage Cart

Remove a selected item or clear the entire cart.

Live Total

The app updates subtotals and the overall total.

Confirm Order

Enter customer name and choose Cash, UPI or Card.

Generate Bill

Save the order and display a receipt with an order ID.

SYSTEM WORKFLOW

1

Launch App

Python starts the Tkinter window.

2

Load Menu

Available food items are read from MySQL.

3

Build Cart

The user selects food and quantities.

4

Review Total

Subtotals and grand total are updated.

5

Confirm

Customer name and payment method are entered.

6

Save & Bill

Order data is saved and the receipt is displayed.

Validation prevents empty-cart orders, missing customer names and invalid quantities.

FOOD MENU & SAMPLE PRICING

ID	FOOD ITEM	CATEGORY	PRICE
1	**Veg Sandwich**	Snacks	₹50
2	**Cheese Sandwich**	Snacks	₹70
3	**Veg Burger**	Fast Food	₹80
4	**French Fries**	Fast Food	₹60
5	**Samosa**	Snacks	₹25
6	**Masala Dosa**	South Indian	₹90
7	**Idli**	South Indian	₹45
8	**Cold Coffee**	Beverages	₹60
9	**Tea**	Beverages	₹15
10	**Water Bottle**	Beverages	₹20

The default menu is inserted when the menu table is empty.

DATABASE DESIGN

MENU_ITEMS

- item_id — Primary key
- name — Food item name
- category — Menu category
- price — Unit price
- available — Availability flag

ORDERS

- order_id — Auto-increment key
- customer_name — Customer
- payment_method — Cash / UPI / Card
- total — Final amount
- order_time — Date and time

ORDER_ITEMS

- order_item_id — Primary key
- order_id — Linked order
- item_id — Menu item ID
- quantity and unit_price
- line_total — Item subtotal

RELATIONSHIP: One order can contain multiple order items.

The program creates the database and tables when needed. Confirmed orders are saved with their item details, and successful transactions are committed.

PYTHON PROGRAM STRUCTURE

CanteenDatabase CLASS

Connects to MySQL, creates tables, loads menu items and saves orders with item details.

CanteenApp CLASS

Builds the Tkinter interface and manages menu display, cart updates, totals, confirmation and receipt display.

DICTIONARIES

DEFAULT_MENU and cart hold item details, prices and quantities.

VALIDATION

Checks selection, positive quantity, non-empty cart and customer name.

ERROR HANDLING

Shows database and file errors using message boxes.

Entry point: main() creates database tables, opens the app window and starts the Tkinter event loop.

SAMPLE OUTPUT — BILL RECEIPT

CANTEEN FOOD ORDERING SYSTEM

Order ID: SAMPLE Customer: Ansh Patel Payment: Cash

Date: Sample date/time

ITEM	QTY	AMOUNT
Veg Burger	2	160.00
Samosa	2	50.00
Tea	1	15.00

TOTAL: ₹225.00

Thank you! Visit again.

Illustrative receipt based on the sample order; ID and date are placeholders.

CONCLUSION & FUTURE SCOPE

CONCLUSION

This project combines Python, Tkinter, OOP, dictionaries and MySQL in a canteen ordering workflow. It supports menu viewing, cart management, automatic totals, order storage and receipt generation.

FUTURE ENHANCEMENTS

- Admin login and menu management
- Food availability updates
- Order history and reports
- Online payment integration
- Customer login and improved printing

KEY LEARNING

A practical demonstration of GUI development, database connectivity, structured programming and data validation in Python.

THANK YOU